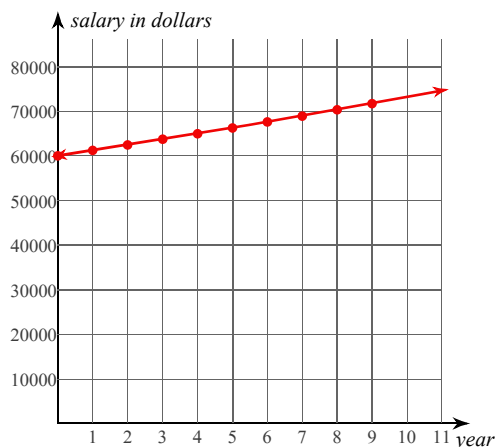


Discrete Exponential Growth and Decay

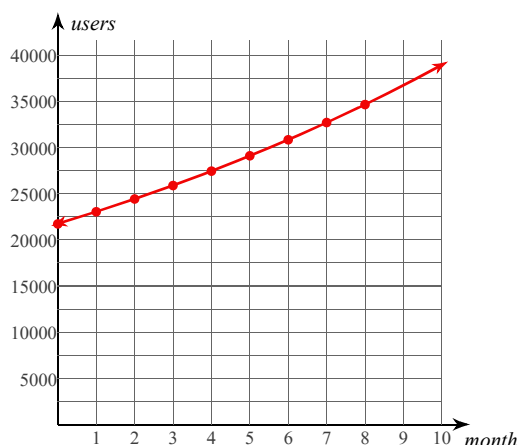
Solve each discrete exponential growth/decay problem. You may use the provided graph to plot points or sketch the exponential function.

- 1) An employee receives a 2% raise once per year. If the employee's initial salary is \$60,000.00, what will the employee's salary be after 9 years?



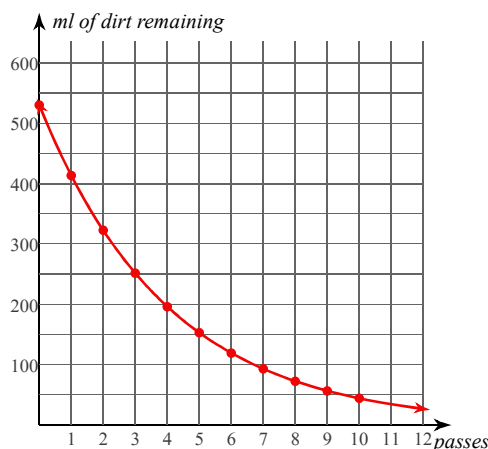
$$60000 \cdot 1.02^9 \approx \$71,705.55$$

- 2) A new social media site is increasing its user base by approximately 6% per month. If the site currently has 21,740 users, what will the approximate user base be 8 months from now?



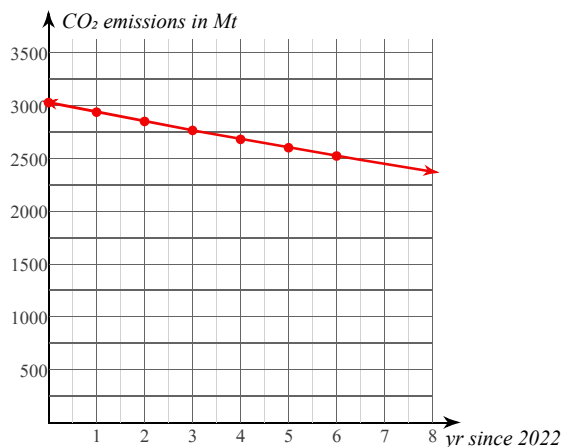
$$21740 \cdot 1.06^8 \approx 34,650 \text{ users}$$

- 3) A robot vacuum cleans a dirty floor using multiple passes. During each pass, 22% of the dirt is removed. If the floor initially has 530.0 ml of dirt, how much dirt will remain after 10 passes?



$$530 \cdot 0.78^{10} \approx 44.2 \text{ ml}$$

- 4) A country pledges to reduce its annual CO₂ emissions by 3% per year. If the emissions in 2022 are 3,030 Mt (metric megatons), what are the maximum allowable emissions in the year 2028?



$$3030 \cdot 0.97^6 \approx 2,524 \text{ Mt}$$